Contents

Education

2023 – Present	PhD in Computational Science, Engineering, and Mathematics Oden Institute, University of Texas at Austin; Austin, TX Advisor: George Biros
2023 – 2026	MS in Computational Science, Engineering, and Mathematics Oden Institute, University of Texas at Austin; Austin, TX
2019 – 2023	BSLAS in Mathematics & Computer Science University of Illinois at Urbana-Champaign; Champaign, IL

Research Experience

Undergraduate Students

2026, January – 2026, May	Max Feldman (UT Austin) PyKokkos GPU kernels for randomized k-nearest neighbors.

Publications

An up-to-date list is available on Google Scholar:
https://scholar.google.com/citations?user=Lmwd5akAAAAJ&hl=en

Refereed Conference & Journal Articles

[A1] **G. Kosmacher**, D. Max, Z. Rapti, C. E. Cáceres, and T. E. Stewart Merrill, "The role of host immunity and the environment in seasonal disease dynamics," *Bulletin of Mathematical Biology*, vol. 88, no. 2, p. 18, Jan. 2026. DOI: 10.1007/s11538-025-01582-3.

[A2] **G. Kosmacher** and K. Shankari, "Evaluating the interplay between trajectory segmentation and mode inference error," *Transportation Research Record: Journal of the Transportation Research Board*, vol. 2678, no. 7, pp. 32–49, Jul. 2024. DOI: 10.1177/03611981231208154.

[A3] Y. Du, **G. Kosmacher**, Y. Liu, J. Massman, J. Palmer, T. Thieme, J. Wu, and Z. Zhang, "Packing densities of delzant and semitoric polygons," *SIGMA. Symmetry, Integrability and Geometry: Methods and Applications*, vol. 19, p. 081, Oct. 2023. DOI: 10.3842/SIGMA.2023.081.

Preprints

[PP1] **G. Kosmacher**, Z. Du, J. Bagge, and G. Biros, "A performance portable fast ewald summation for stokes flow," Jun. 2026. arXiv: 2606.19059 [math.NA].

[PP2] **G. Kosmacher**, E. T. Phipps, and S. Rajamanickam, "A performance portable matrix free dense mttkrp in genten," preprint, Oct. 2025. arXiv: 2510.14891 [cs.MS].

Talks

[T1] **G. Kosmacher**. "Performance portable fast ewald summation." (May 2025), [Online]. Available: https://youtu.be/gZ2Pi5LSP6M?feature=shared.

Software

My open-source contributions are from the kennykos *GitHub account.*

ut-padas/parki	Project maintainer and lead contributor. Developed CPU/GPU Ewald summation algorithms for the Stokes and Laplace kernels in different periodic regimes.
kokkos/pykokkos	Project collaborator and top 10 contributor. Implemented functionality for mathematical special functions, automatic just-in-time (JIT) recompilation, CMake JIT compilation, building and installing on MacOS and AMD GPUs, documentation and continuous integration.
sandialabs/GenTen	Implemented a matrix-free MTTKRP for dense tensors: MTTKRP_Dense_Perm_Kernel, the basis for the publication *A Performance Portable Matrix Free Dense MTTKRP in GenTen*.